

StockPilot - Full Documentation

1. Application Architecture

StockPilot is a Next.js-based inventory management system using Supabase as backend.

```
Frontend: Next.js (App Router)
Backend: Supabase (Database + API)
Styling: Tailwind CSS
State: React Hooks
```

2. Folder Structure

```
src/
├── app/
│   ├── components/
│   │   ├── Navbar.js
│   │   └── Logo.js
│   ├── dashboard/page.js
│   └── layout.js
├── lib/
│   ├── supabase.js
│   └── stock.js
├── config/
└── company.js
```

3. Database Structure

Products Table

```
products
id
name
minimum_stock
cost_price
selling_price
created_at
```

Production Table

```
production
id
product_id
quantity
created_at
```

Dispatch Table

```
dispatch
id
```

```
product_id
customer_id
quantity
created_at
```

4. Stock Calculation Logic

```
Starting Stock
+ Production
- Dispatch
= Current Stock
```

5. Supabase Connection

```
import { createClient } from '@supabase/supabase-js';

export const supabase = createClient(
  process.env.NEXT_PUBLIC_SUPABASE_URL,
  process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY
);
```

6. Stock Utility

```
export async function getCurrentStock(product){
  const { data: production } = await supabase
    .from("production")
    .select("quantity")
    .eq("product_id", product.id);

  const { data: dispatch } = await supabase
    .from("dispatch")
    .select("quantity")
    .eq("product_id", product.id);

  const totalProduction = production.reduce((s,x)=>s+x.quantity,0);
  const totalDispatch = dispatch.reduce((s,x)=>s+x.quantity,0);

  return totalProduction - totalDispatch;
}
```

7. Layout System

```
export default function RootLayout({ children }) {
  return (
    <html>
      <body className="flex">
        <Navbar />
        <main className="ml-64 p-6 w-full">
          {children}
        </main>
      </body>
    </html>
  );
}
```

8. Navbar Component

```
export default function Navbar(){
  return (
    <aside className="w-64 h-screen fixed left-0 top-0 border-r">
      <div className="p-4 font-bold">StockPilot</div>
    </aside>
  );
}
```

9. Dashboard Logic

```
const [totalProducts,setTotalProducts]=useState(0);
const [totalStock,setTotalStock]=useState(0);
```

Dashboard loads products, calculates stock, and displays alerts + activity.